

Show understanding of the need for and examples of the use of compression

9608_s21_qp_11_sc_8.b.i

QUESTION

(b) The game generates a story which is stored as a text file.
Jay compresses the text file using lossless compression before sending it by email to his friend.

(i) Identify **two** reasons for compressing the text file.

1

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2

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[2]

MARK SCHEME

8(b)(i)	1 mark per bullet point to max 2 - To reduce the storage space required to store the file - To reduce the time it would take to transmit the file - Because they have to email it and have limited file sizes for sending/receiving	2
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9608_s19_qp_11_sc_1.d.i

QUESTION

(d) A recording of a concert is stored as a file. The file is compressed using lossy compression before it is streamed to users.

(i) State why this file needs to be compressed.

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[1]

MARK SCHEME

1(d)(i)	1 mark per bullet point to max 1 The data files are very large It would take a long time to send the uncompressed file // Compressed files will download faster A higher bandwidth would be needed to transmit the uncompressed file	1
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9608_w16_qp_12_sc_2.c.ii

QUESTION

- (ii) Give **one** file technique that could be applied when either of the two features, described in **part (c)(i)**, are present.

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.....[1]

MARK SCHEME

(ii) (File) <u>compression</u>	[1]
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Show understanding of lossy and lossless compression and justify the use of a method in a given situation

9608_w21_qp_11_sc_1.c

QUESTION

(c) A bitmap image needs to be compressed before it can be sent by email.

Describe **one** lossy and **one** lossless method of compressing the image.

Lossy

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Lossless

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[4]

MARK SCHEME

1(c)	One mark per bullet point to max 2 for each method. Lossy • Reduce the resolution • ...fewer pixels per unit measurement • ...fewer pixels / binary numbers are stored Lossless • RLE (Run Length Encoding) • ...looks for runs of consecutive pixels of the same colour • ...stores the colour value once and the number of times it occurs	4
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QUESTION

(c) The bitmap images are compressed for use on a website.

Tick (✓) **one** box to select the **most appropriate** type of compression for the images used on the website and justify your answer.

Lossy	☐
Lossless	☐

Justification

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..... [3]

MARK SCHEME

7(c)	Justification 1 mark per bullet point to max 3 Lossy: • Lossy compression will achieve a much higher compression ratio (than lossless) for photographic images... • ...so the images will load faster • There will be little identifiable change to the viewed images • It may not be important that the originals are not recoverable Lossless: • Images may be required in their original form // may need to be recovered completely • Images may already be a small file size so major reduction in file size is not needed... • ...as the files will load quickly	3
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9608_s21_qp_11_sc_8.b.ii

QUESTION

- (ii) Explain the reasons why Jay compresses the text file with lossless compression instead of lossy compression.

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..... [2]

MARK SCHEME

8(b)(ii)	1 mark per bullet point	2
	- All of the original data needs to be retained/recoverable ... because otherwise the file contents would not make sense	

9608_s20_qp_13_sc_5.b

QUESTION

- (b) Wei wants to compress the source code to transport it to another computer.

Identify the **most appropriate** compression technique he should use.

Justify your choice.

Compression technique

Justification

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.....

[3]

MARK SCHEME

5(b)	1 mark for a lossless technique 1 mark per bullet point for justification to max 2	3
	- Lossless does not lose any data - Any lost data will mean the program will not work	

QUESTION

- (b) The file is too large to be emailed and the file size needs to be reduced.
- (i) Name **one** lossless compression technique that can be used to reduce the size of the sound file.

.....

..... [1]

MARK SCHEME

2(b)(i)	1 mark only from: • Run-length encoding • Huffman Coding • Any other valid encoding method	1
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QUESTION

- (ii) Describe **one** lossy compression technique that can be used to reduce the size of the sound file.

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..... [2]

MARK SCHEME

2(b)(ii)	1 mark per bullet point to max 2 • Reduce sampling rate • ... fewer samples taken per second means less data is being stored Or • Reduce sample resolution • ... fewer bits are used to represent each sample, so less data is stored	2
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QUESTION

(b) File compression is one example of a utility program.

Tick (✓) **one** box on each row to indicate whether the action is an example of lossy compression or lossless compression.

Action	Lossy	Lossless
Reducing the resolution of an image		
Using run-length encoding on a text file		
Reducing the sampling rate of a sound file		

[1]

MARK SCHEME

9(b)	1 mark for 3 correct answers	1												
	<table> <tr> <th>Action</th><th>Lossy</th><th>Lossless</th></tr> <tr> <td>Reducing the resolution of an image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding on a text file</td><td></td><td>✓</td></tr> <tr> <td>Reducing the sampling rate of a sound file</td><td>✓</td><td></td></tr> </table>	Action	Lossy	Lossless	Reducing the resolution of an image	✓		Using run-length encoding on a text file		✓	Reducing the sampling rate of a sound file	✓		
Action	Lossy	Lossless												
Reducing the resolution of an image	✓													
Using run-length encoding on a text file		✓												
Reducing the sampling rate of a sound file	✓													

QUESTION

(e) Data can be compressed using either lossy or lossless compression.

Tick (✓) **one** box in each scenario to identify whether lossy or lossless compression should be used. Justify your choice.

(i) A program written in a high-level language.

Lossy	Lossless

Justification

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..... [2]

MARK SCHEME

5(e)(i)	1 mark per bullet to max 2 Lossless: All the data is needed // the original file is fully restored If any data is lost, the program will not run Probably does not require significant reduction in file size // a program written in a high-level language is just text, so does not need much reduction in size	2
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9608_w19_qp_11_sc_5.e.ii

QUESTION

(ii) A photograph that needs to be emailed to a friend.

Lossy	Lossless

Justification

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..... [2]

MARK SCHEME

5(e)(ii)	1 mark per bullet to max 2 Lossy: All the data is not required The number of colours / resolution can be reduced without the user noticing Email requires a significantly smaller file size // takes less time to transmit Lossless: A high quality image may be needed All of the data is needed // cannot afford to lose any data // the original file is fully restored	2
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QUESTION

(iii) You need to upload a video that you have created to a website.

Lossy	Lossless

Justification

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..... [2]

MARK SCHEME

5(e)(iii)	1 mark per bullet to max 2 Lossy: Some loss of quality will not be noticed // high quality video not needed on the website A more significant reduction may be needed Takes a shorter time to upload / download // requires less bandwidth Lossless: A high quality video may be needed Might only be a short video clip All of the data is needed // cannot afford to lose any data // the original file is fully restored	2
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QUESTION

(ii) Define the term **lossy compression**.

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..... [1]

MARK SCHEME

1(d)(ii)	1 mark per bullet point to max 1 Data is lost The decompressed file is not the same as the original	1
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QUESTION

- (iii) The file could be compressed using lossless compression.

Explain why lossy compression is a more appropriate compression technique than lossless for this file.

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..... [3]

MARK SCHEME

1(d)(iii)	1 mark per bullet point to max 3 Lossy creates a smaller file than lossless // lossy compresses further than lossless The recording of the concert is a large file size and needs <u>significant</u> reduction in size Lossy removes detail which can be lost without people noticing By example e.g. reduction in sound quality <u>will not be noticed</u>	3
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QUESTION

- (ii) The photograph needs to be sent by email but the file size is too big. It needs to be compressed.

The table lists several methods of making an image file size smaller.

Tick (✓) **one** box on each row to indicate whether each method is lossy or lossless.

Compression method	Lossy	Lossless
Cropping the image		
Reducing the resolution of the image		
Using run-length encoding (RLE)		
Reducing the colour depth of the image		

[4]

MARK SCHEME

1(b)(ii)	1 mark for correct tick in each row.	4															
<table border="1"> <thead> <tr> <th>Compression method</th><th>Lossy</th><th>Lossless</th></tr> </thead> <tbody> <tr> <td>Cropping the image</td><td>✓</td><td></td></tr> <tr> <td>Reducing the resolution of the image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding (RLE)</td><td></td><td>✓</td></tr> <tr> <td>Reducing the colour depth of the image</td><td>✓</td><td></td></tr> </tbody> </table>			Compression method	Lossy	Lossless	Cropping the image	✓		Reducing the resolution of the image	✓		Using run-length encoding (RLE)		✓	Reducing the colour depth of the image	✓	
Compression method	Lossy	Lossless															
Cropping the image	✓																
Reducing the resolution of the image	✓																
Using run-length encoding (RLE)		✓															
Reducing the colour depth of the image	✓																

9608_w18_qp_12_sc_1.b.ii

QUESTION

(ii) The file size of the photograph needs to be reduced before it is placed on the website.

Draw lines to link each method of reducing the file size of the image to:

- its description and
- its compression type, where appropriate.

Description	Method	Compression type
Removes pixels	Crop the photograph	Lossy
Reduces number of pixels per inch	Use run-length encoding	Lossless
Uses fewer bits per pixel	Use fewer colours	
Stores colour code and count of repetitions		

[5]

MARK SCHEME

1(b)(ii)	1 mark per method correctly linked to its description max 3 1 mark for each compression type correctly linked to its method(s). max 2	5																		
	<table border="1"> <thead> <tr> <th>Description</th><th>Method</th><th>Compression type</th></tr> </thead> <tbody> <tr> <td>Removes pixels</td><td>Crop the photograph</td><td>Lossy</td></tr> <tr> <td>Reduces number of pixels per inch</td><td>Crop the photograph</td><td>Lossy</td></tr> <tr> <td>Uses fewer bits per pixel</td><td>Use run-length encoding</td><td>Lossless</td></tr> <tr> <td>Stores colour code and count of repetitions</td><td>Use run-length encoding</td><td>Lossless</td></tr> <tr> <td></td><td>Use fewer colours</td><td>Lossy</td></tr> </tbody> </table>	Description	Method	Compression type	Removes pixels	Crop the photograph	Lossy	Reduces number of pixels per inch	Crop the photograph	Lossy	Uses fewer bits per pixel	Use run-length encoding	Lossless	Stores colour code and count of repetitions	Use run-length encoding	Lossless		Use fewer colours	Lossy	
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Stores colour code and count of repetitions	Use run-length encoding	Lossless																		
	Use fewer colours	Lossy																		

9608_s16_qp_12_sc_4.b.i

QUESTION

(b) The students use software to compress the sound clips before emailing them.

(i) Circle your chosen method of compression and justify your choice.

Lossy / Lossless

Justification:

[3]

MARK SCHEME

(b) (i) **No mark** awarded for identifying method. **Three marks** for justification.

[3]

Lossy – Three points from:

- The human ear will not notice that the decompressed stream will not be identical to the original (file) / that parts of the original data have been discarded / removed / deleted.
- File size reduction is greater than using lossless.
- Email has limits on file sizes (on attachments) / a smaller file will take less time to transmit.
- The file may not need to be of high precision / accuracy.
- The producer has requested an mp3 file.

Lossless – Three points from:

- The file needs to be high precision / accuracy.
- None of the original data is lost / the decompressed file will be identical to the original.
- The producer has requested a flac file.

QUESTION

(b) The students use software to compress the sound clips before emailing them.

(i) Circle your chosen method of compression and justify your choice.

Lossy / Lossless

Justification:

.....

.....

.....[3]

MARK SCHEME

(b) (i) **No mark** awarded for identifying method. **Three marks** for justification.

[3]

Lossy – Three points from:

- The human ear will not notice that the decompressed stream will not be identical to the original (file) / that parts of the original data have been discarded / removed / deleted.
- File size reduction is greater than using lossless.
- Email has limits on file sizes (on attachments) / a smaller file will take less time to transmit.
- The file may not need to be of high precision / accuracy.
- The producer has requested an mp3 file.

Lossless – Three points from:

- The file needs to be high precision / accuracy.
- None of the original data is lost / the decompressed file will be identical to the original.
- The producer has requested a flac file.

QUESTION

(c) Explain the difference between *lossless* and *lossy* data compression techniques.

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.....[3]

MARK SCHEME

(c) Any **three** from:

For full marks both techniques must be mentioned.

- lossless designed to lose none of the original detail/lossless allows original file to be recreated exactly
- lossless technique based on some form of replacement
- mention of type of replacement, for example RLE, FLAC etc.
- by example: e.g. 000–1111–222222–333 = 3–0, 4–1, 6–2, 3–3 etc.
- maximum compression about 50%
- lossy may result in loss of detail compared to original file/lossy does not allow original file to be re-created exactly
- lossy techniques make decision about what parts of sound/sound file are important and discards other information
- only keeps sounds human ear can process/discards sounds most people cannot hear
- ... then applies lossless technique, for further reduction
- lossy compression can reduce to about 10%
- an example of jpeg, mp3 or other correct examples of compressed formats.

No double credit to opposite answers, e.g. lossless maintains detail, but lossy loses detail just one mark.

[3]

9608_s15_qp_11_sc_2.c

QUESTION

(c) Explain the difference between *lossless* and *lossy* data compression techniques.

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.....[3]

MARK SCHEME

(c) Any three from:

For full marks both techniques must be mentioned.

- lossless designed to lose none of the original detail/lossless allows original file to be recreated exactly
- lossless technique based on some form of replacement
- mention of type of replacement, for example RLE, FLAC etc.
- by example: e.g. 000–1111–222222–333 = 3–0, 4–1, 6–2, 3–3 etc.
- maximum compression about 50%
- lossy may result in loss of detail compared to original file/lossy does not allow original file to be re-created exactly
- lossy techniques make decision about what parts of sound/sound file are important and discards other information
- only keeps sounds human ear can process/discards sounds most people cannot hear
- ... then applies lossless technique, for further reduction
- lossy compression can reduce to about 10%
- an example of jpeg, mp3 or other correct examples of compressed formats.

No double credit to opposite answers, e.g. lossless maintains detail, but lossy loses detail just one mark.

[3]

Show understanding of how a text file, bitmap image, vector graphic and sound file can be compressed

9608_w21_qp_11_sc_1.c

QUESTION

(c) A bitmap image needs to be compressed before it can be sent by email.

Describe **one** lossy and **one** lossless method of compressing the image.

Lossy

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Lossless

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[4]

MARK SCHEME

1(c)	One mark per bullet point to max 2 for each method. Lossy • Reduce the resolution • ...fewer pixels per unit measurement • ...fewer pixels / binary numbers are stored • Reduce the colour depth • ...reduce the number of bits per colour • ...each pixel has fewer bits Lossless • RLE (Run Length Encoding) • ...looks for runs of consecutive pixels of the same colour • ...stores the colour value once and the number of times it occurs	4
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QUESTION

- (b) Videos on the website are compressed. Compression algorithms can use spatial redundancy or temporal redundancy.

Describe **spatial redundancy** and **temporal redundancy**.

Spatial

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Temporal

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[4]

MARK SCHEME

9(b)	1 mark per bullet point, max 2 for each Spatial Redundancy • Intra-frame // Redundancy within a frame • Pixels in a single video frame have the same value • A description of an appropriate compression method e.g. RLE Temporal Redundancy • Inter-frame // Redundancy between frames • Pixels in a sequence of consecutive video frames have the same value in the same location • ...records only the differences between the frames	4
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QUESTION

- (b) The file is too large to be emailed and the file size needs to be reduced.

- (i) Name **one** lossless compression technique that can be used to reduce the size of the sound file.

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..... [1]

MARK SCHEME

2(b)(i)	1 mark only from: - Run-length encoding - Huffman Coding - Any other valid encoding method	1
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9608_s20_qp_12_sc_2.b.ii

QUESTION

(ii) Describe **one** lossy compression technique that can be used to reduce the size of the sound file.

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..... [2]

MARK SCHEME

2(b)(ii)	1 mark per bullet point to max 2 - Reduce sampling rate ... fewer samples taken per second means less data is being stored Or - Reduce sample resolution ... fewer bits are used to represent each sample, so less data is stored	2
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9608_s20_qp_11_sc_9.b

QUESTION

(b) File compression is one example of a utility program.

Tick (✓) **one** box on each row to indicate whether the action is an example of lossy compression or lossless compression.

Action	Lossy	Lossless
Reducing the resolution of an image		
Using run-length encoding on a text file		
Reducing the sampling rate of a sound file		

[1]

MARK SCHEME

9(b)	1 mark for 3 correct answers	1												
<table border="1"> <thead> <tr> <th>Action</th><th>Lossy</th><th>Lossless</th></tr> </thead> <tbody> <tr> <td>Reducing the resolution of an image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding on a text file</td><td></td><td>✓</td></tr> <tr> <td>Reducing the sampling rate of a sound file</td><td>✓</td><td></td></tr> </tbody> </table>			Action	Lossy	Lossless	Reducing the resolution of an image	✓		Using run-length encoding on a text file		✓	Reducing the sampling rate of a sound file	✓	
Action	Lossy	Lossless												
Reducing the resolution of an image	✓													
Using run-length encoding on a text file		✓												
Reducing the sampling rate of a sound file	✓													

9608_w18_qp_13_sc_1.b.ii

QUESTION

- (ii) The photograph needs to be sent by email but the file size is too big. It needs to be compressed.

The table lists several methods of making an image file size smaller.

Tick (✓) **one** box on each row to indicate whether each method is lossy or lossless.

Compression method	Lossy	Lossless
Cropping the image		
Reducing the resolution of the image		
Using run-length encoding (RLE)		
Reducing the colour depth of the image		

[4]

MARK SCHEME

1(b)(ii)	1 mark for correct tick in each row.	4															
<table border="1"> <thead> <tr> <th>Compression method</th><th>Lossy</th><th>Lossless</th></tr> </thead> <tbody> <tr> <td>Cropping the image</td><td>✓</td><td></td></tr> <tr> <td>Reducing the resolution of the image</td><td>✓</td><td></td></tr> <tr> <td>Using run-length encoding (RLE)</td><td></td><td>✓</td></tr> <tr> <td>Reducing the colour depth of the image</td><td>✓</td><td></td></tr> </tbody> </table>			Compression method	Lossy	Lossless	Cropping the image	✓		Reducing the resolution of the image	✓		Using run-length encoding (RLE)		✓	Reducing the colour depth of the image	✓	
Compression method	Lossy	Lossless															
Cropping the image	✓																
Reducing the resolution of the image	✓																
Using run-length encoding (RLE)		✓															
Reducing the colour depth of the image	✓																

QUESTION

(c) Explain how run-length encoding would compress the image in **part (a)**.

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.....[3]

MARK SCHEME

1(c)	1 mark per bullet point to max 3 - Looks for runs of consecutive pixel of the same colour - Stores the colour value once and the number of times it occurs - Lossless method of compression - Reference to the given image in context	3
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QUESTION

(ii) The file size of the photograph needs to be reduced before it is placed on the website.

Draw lines to link each method of reducing the file size of the image to:

- its description and
- its compression type, where appropriate.

Description	Method	Compression type
Removes pixels	Crop the photograph	Lossy
Reduces number of pixels per inch	Use run-length encoding	Lossless
Uses fewer bits per pixel	Use fewer colours	
Stores colour code and count of repetitions		

[5]

MARK SCHEME

1(b)(ii)	1 mark per method correctly linked to its description max 3 1 mark for each compression type correctly linked to its method(s). max 2 Description Removes pixels Reduces number of pixels per inch Uses fewer bits per pixel Stores colour code and count of repetitions Method Crop the photograph Use run-length encoding Use fewer colours Compression type Lossy Lossless	5
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9608_s18_qp_13_sc_6.b

QUESTION

(b) Run-length encoding (RLE) is used to store the image with the following colour codes.

Colour	Code
Black	1A
White	3B

Show how run-length encoding is used to store the image.

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[3]

MARK SCHEME

6(b)	1 mark for correct method (colour code and number of pixels) 1 mark for first 7 groups correct 1 mark for remainder correct 3B9 1A3 3B3 1A2 3B1 1A2 3B2 1A1 3B3 1A1 3B2 1A2 3B1 1A2 3B3 1A3 3B9	3
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9608_s18_qp_11_sc_2.b.ii

QUESTION

- (ii) The image is compressed before it is attached to an email.

Explain how run-length encoding (RLE) will compress the image.

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.....[2]

MARK SCHEME

2(b)(ii)	1 mark for the explanation Stores the colour and the number of times it occurs 1 mark for example from An example from the bitmap given e.g. B5, W1, B1 and so on	2
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9608_s16_qp_12_sc_4.b.ii

QUESTION

Students also email images to the radio station for use on its website.

These are compressed before sending using run-length encoding (RLE).

- (ii) Explain what is meant by run-length encoding.

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.....[3]

MARK SCHEME

(ii) Three points from:

[3]

- Lossless method of compression.
- Reduces (the physical size of) a string of adjacent, identical characters/pixels / bytes etc..
- The repeating string (a run) is encoded into two values.
- One value represents the number of (identical) characters in the run (the run count).
- The other value is the code of the character / colour code of pixel etc. in the run (the run value).
- The run value and run count combination may be preceded by a control character.
- Any valid example given.

9608_s16_qp_12_sc_4.b.iii

QUESTION

(iii) The following diagrams show:

- the denary colour code that represents each colour
- the first three rows of a bitmap image

Colour symbol	Colour code (denary)
B	153
W	255

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	B	B	B	B	B	B	B	B	B	B	W	W	W	B	B	B
1	B	B	B	B	B	B	B	B	B	W	W	W	W	W	W	B
2	B	B	B	B	B	B	B	W	W	W	W	W	W	W	W	W
...																
95																

Show how RLE will compress the first three rows of this image.

Row 1:

Row 2:

Row 3:[2]

MARK SCHEME

(iii) **Two marks for three correct rows, one mark for two correct rows.**

[2]

Row 1: 153 10 255 3 153 3

Row 2: 153 9 255 6 153 1

Row 3: 153 7 255 9

Alternative correct answer:

Row 1: 153 9 255 2 153 2

Row 2: 153 8 255 5 153 0

Row 3: 153 6 255 8

9608_s16_qp_11_sc_4.b.ii

QUESTION

Students also email images to the radio station for use on its website.

These are compressed before sending using run-length encoding (RLE).

(ii) Explain what is meant by run-length encoding.

.....

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

(ii) **Three points from:**

[3]

- Lossless method of compression.
- Reduces (the physical size of) a string of adjacent, identical characters/pixels / bytes etc..
- The repeating string (a run) is encoded into two values.
- One value represents the number of (identical) characters in the run (the run count).
- The other value is the code of the character / colour code of pixel etc. in the run (the run value).
- The run value and run count combination may be preceded by a control character.
- Any valid example given.

QUESTION

(iii) The following diagrams show:

- the denary colour code that represents each colour
- the first three rows of a bitmap image

Colour symbol	Colour code (denary)
B	153
W	255

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	B	B	B	B	B	B	B	B	B	B	W	W	W	B	B	B
1	B	B	B	B	B	B	B	B	B	W	W	W	W	W	W	B
2	B	B	B	B	B	B	B	W	W	W	W	W	W	W	W	W
...																
95																

Show how RLE will compress the first three rows of this image.

Row 1:

Row 2:

Row 3:[2]

MARK SCHEME

(iii) **Two marks** for **three** correct rows, **one mark** for **two** correct rows.

[2]

Row 1: 153 10 255 3 153 3

Row 2: 153 9 255 6 153 1

Row 3: 153 7 255 9

Alternative correct answer:

Row 1: 153 9 255 2 153 2

Row 2: 153 8 255 5 153 0

Row 3: 153 6 255 8